Dr. Thyagarajulu Gollapalli

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Research Interests

Computational geodynamics; geophysics; subduction dynamics; megathrust seismicity; mantle convection; fault representation in geodynamic models; numerical modelling; high-performance computing; reproducible open-source Earth-science workflows; geophysical data acquisition, processing, and interpretation.

Current and Past Work Experience

Research Officer/Software Engineer Jan. 2026–Present
Australian National University *Canberra, ACT*

- Open-source developer for the Underworld geodynamics codebase, contributing core features, user-facing interfaces, testing, documentation, and reproducible modelling workflows.

Assistant Lecturer, Maternity Cover Jun. 2025–Dec. 2025
Monash University *Melbourne, VIC*

Senior Research Officer/Software Engineer Aug. 2023–Jun. 2025
Monash University and AuScope *Melbourne, VIC*

Imaging Geophysicist Nov. 2022–Jun. 2023
CGG *Perth, WA*

Junior Research Fellow in Gravity and Magnetics Lab May 2017–Apr. 2018
Department of Earth Sciences, IIT Bombay *Mumbai, India*

Project Assistant in Computational Geodynamics Lab Aug. 2015–Mar. 2017
Centre for Earth Sciences, IISc *Bengaluru, India*

Education

PhD, Computational Geodynamics and Geophysics Apr. 2018–Nov. 2022
Monash University and Indian Institute of Technology Bombay
Thesis: *Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling*

BSc, Mathematics; Minor in Earth and Environmental Science Aug. 2011–May 2015
Indian Institute of Science Bengaluru, India

Top Three Publications

1. Capitanio, F. A., **Gollapalli, T.**, Munukutla, R., Graciosa, J. C., Zuhair, M., Beall, A., & Dal Zilio, L. (2025). Bridging the gap between subduction dynamics and the long-term strength of megathrusts. *Nature Communications*. doi:10.1038/s41467-025-65824-7.
2. Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T.**, Lu, N., & Beucher, R. (2025). Underworld3: Mathematically self-describing modelling in Python for desktop, HPC and cloud. *Journal of Open Source Software*. doi:10.21105/joss.07831.
3. Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T.**, Tang, T., & Zuhair, M. (2025). Testing driving mechanisms of megathrust seismicity with explainable artificial intelligence. *JGR: Solid Earth*. doi:10.1029/2024JB028774.

Awards

- Faculty of Science Dean's International Postgraduate Research Scholarship, Monash University, 2018–2022.
- Monash Graduate Scholarship, Monash University, 2018–2021.
- Postgraduate Publications Award, Monash University, 2022.
- AGU Fall Meeting Student Travel Grant, 2021.
- Junior Research Fellow Scholarship, Ministry of Human Resource Development, India, 2017–2018.
- INSPIRE Scholarship for Higher Education, Department of Science and Technology, India, 2011–2015.